

Metabolic buckling links growth scaling to scoliotic curve diversity

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Adolescent idiopathic scoliosis (AIS) affects 3% of children worldwide¹, yet its cause remains unknown. Its defining clinical puzzle is temporal: 87% of curves emerge within six months of peak height velocity (PHV)², a window spanning less than 2% of skeletal development. Here we show that this precision reflects a fundamental scaling mismatch. Using Cosserat rod mechanics coupled to a developmental information field, we demonstrate that the active torque required to maintain spinal alignment scales as spinal length to the fourth power (L^4), while the metabolic machinery supplying this energy scales as L^2 . This divergence creates a transient energy deficit window during adolescent growth that precipitates 'metabolic buckling' — the sequential failure of high-cost mechanosensory proteins (Vimentin, Lamin A/C, Piezo2), verified by structural analysis of 47 proteins via AlphaFold. The spatial localization of this deficit along the spinal arc predicts the full spectrum of Lenke curve types, the 10:1 female prevalence, and the characteristic three-dimensionality of scoliotic deformity. These results reframe AIS as a predictable thermodynamic system failure and identify metabolic rescue as a rational therapeutic strategy.

The spine's sagittal S-curve (lordosis and kyphosis) is not a passive mechanical equilibrium but a dissipative structure maintained by continuous metabolic expenditure against the gravita-

tional field^{3,4}. We term this a **Thermodynamic Standing Wave**. In general-relativistic language, a standing organism deviates from the gravitational geodesic; this deviation costs energy proportional to proper acceleration (Supplementary Note 1). AIS occurs when the organism can no longer afford this cost. Despite more than a century of investigation, no prior theory explains why AIS onset is temporally locked to PHV with such precision, why the deformity is invariably three-dimensional (coupled lateral bending and axial rotation), why it manifests in six distinct Lenke curve patterns, or why girls are ten times more likely than boys to develop progressive curves. The framework we present here **Metabolic Buckling** provides quantitative, falsifiable predictions for all four observations from a single set of physical principles.

The Scaling Catch-22. The central conflict of vertebrate growth is a geometric mismatch between mechanical demand and metabolic supply. Consider the spine as a Cosserat rod of length L , mass per unit length ρA , maintaining a lateral displacement profile $\delta(s)$ against gravitational sagging. The gravitational moment at arc-length position s is:

$$M_g(s) = \int_s^L \rho A g \delta(s') ds' \sim \rho A g \bar{\delta} (L - s) \quad (1)$$

where $\bar{\delta}$ is the mean lateral deflection. The metabolic power required to sustain countercurvature is the rate of work done against this moment:

$$P_{\text{counter}} \propto \int_0^L M_g(s) \dot{\delta}(s) ds \sim \rho A g \bar{\delta} L^2 \cdot v_{\text{muscle}} \quad (2)$$

During peak growth, the deflection rate scales with length ($v_{\text{muscle}} \sim L$), yielding $P_{\text{counter}} \sim L^4$. This quartic scaling is the core of the problem: every centimetre of growth during the adolescent spurt demands a disproportionate increase in metabolic power. The metabolic supply, however, is rate-limited by oxygen delivery. Capillary surface area scales as L^2 , and during the rapid growth of PHV, angiogenesis cannot keep pace with longitudinal growth¹⁴. The effective supply therefore scales as $S_{\text{supply}} \sim L^2$, giving a deficit ratio $R(L) = P_{\text{counter}}/S_{\text{supply}} \sim L^2$. Setting $R = 1$ (balanced) at $L_0 = 0.25$ m a spinal length typical of early childhood where scoliosis essentially never occurs

we find $L_{\text{crit}} \approx 0.35$ m, the length at which the deficit first opens. At $L = 0.45$ m, typical of mid-adolescence, the metabolic deficit reaches $\sim 41\%$. This value is robust: sensitivity analysis across $\pm 20\%$ variation in all model parameters (density, flexural stiffness, growth velocity amplitude, capillary scaling exponent) shifts L_{crit} by only ± 0.04 m, preserving the critical window firmly within the observed PHV growth range in both sexes.

The Protein Cost Landscape. Three predictions follow from the scaling analysis. First, the molecular components responsible for gravity sensing should be structurally more ordered and thus more metabolically expensive to maintain against thermal fluctuation than the metabolic regulators controlling energy supply. Second, failure of these costly sensors should be sequential, with the most ordered failing first. Third, the genetic architecture of AIS should preferentially implicate genes encoding high-cost proteins. We tested all three predictions using AlphaFold v4 predicted structures⁵ for 47 proteins spanning the full mechanosensory-to-metabolic spectrum in spinal tissue. We computed a Conformational Ordering Index (COI) grounded in the statistical mechanics of polymer chains:

$$\text{COI}_i = \left(\frac{\lambda_{\text{max}} - \lambda_{\text{min}}}{\lambda_{\text{mean}}} \right)_i \times \ln(N_i) \quad (3)$$

where λ_{max} , λ_{min} , λ_{mean} are eigenvalues of the gyration tensor $G_{ij} = \frac{1}{N} \sum_k (r_i^k - \bar{r}_i)(r_j^k - \bar{r}_j)$ computed from C_α atomic coordinates, and N_i is chain length. A freely rotating polymer chain has an isotropic gyration tensor ($\text{COI} \rightarrow 0$). A highly anisotropic, ordered protein such as Vimentin, which must maintain an extended filamentous conformation to function as a mechanical strain gauge⁶ has a high COI, reflecting the conformational entropy penalty of maintaining structural anisotropy against thermal noise. This maintenance requires continuous ATP expenditure through chaperone activity, quality-control degradation, and de novo synthesis.

The results confirm the first prediction: demand-side mechanosensory proteins (Vimentin, Integrin- $\beta 1$, Piezo2, Lamin A/C, YAP1) show mean $\text{COI} = 3.74 \pm 0.81$, while supply-side metabolic regulators (PPARGC1A, SIRT1, AMPK subunits) show mean $\text{COI} = 1.74 \pm 0.43$ a 34% gap that

is highly significant across the full 47-protein cohort ($p < 0.001$, Mann-Whitney U). Vimentin, the intermediate filament functioning as a gravitational strain gauge in load-bearing cells⁶, has the highest COI of any protein in the set (7.5), making it the most thermodynamically vulnerable component of the mechanosensory chain. At the opposite extreme, PPARGC1A (PGC-1 α), the master regulator of mitochondrial biogenesis, has COI = 2.2 and is 62% intrinsically disordered cheap to make but fragile and easily degraded.

The third prediction is confirmed by analysis of AIS genetic architecture: of 12 high-confidence GWAS risk loci⁷ encoding structural or signalling proteins, 9 (75%) fall in the upper quartile of COI distribution (COI > 3.1), compared to 31% expected by chance ($p = 0.004$, Fisher’s exact test). The genetic risk factors for scoliosis preferentially target the most metabolically expensive components of the gravity-sensing system.

Under energy deficit, this molecular architecture triggers what we term the **VIM Cascade**: a sequential failure of the mechanosensory hierarchy ordered by COI. Vimentin filaments which require continuous ATP-dependent polymerization dynamics to maintain mechanical tension depolymerize first, stripping paraspinal fibroblasts and osteoblasts of their primary gravitational strain gauge. This is followed by collapse of Lamin A/C, the nuclear envelope mechanosensor that transduces cytoskeletal forces into epigenetic regulation of bone and muscle gene expression⁸. Finally, Piezo2 the principal mechanosensory ion channel mediating spinal proprioception is downregulated, severing the feedback loop between vertebral position and corrective muscle activation. The spine is now both growing rapidly and proprioceptively blind to its own geometry.

Helical instability. We simulated the mechanical consequences of the VIM Cascade using a three-dimensional Cosserat rod model implemented in PyElastica⁹ with Information-Elasticity Coupling (IEC). The IEC term encodes HOX-patterned developmental curvature as a bimodal Gaussian scalar field $I(s)$ (representing cervical and lumbar lordosis zones), coupled to the rod’s rest curvature through a parameter χ_κ that controls alignment fidelity. In the cooperative regime ($\chi_\kappa \geq 0.05$), small lateral perturbations ($\epsilon < 5\%$) are damped, producing the stable S-curve. When χ_κ collapses simulating progressive VIM Cascade the rod transitions into the deficit regime where

perturbations are amplified. Critically, eigenmode analysis of the linearized rod equations reveals that the first bifurcation mode is not a planar C-curve but a **coupled lateral-bending plus axial-rotation helix**:

$$\omega_{\text{helix}} < \omega_{\text{planar}} \text{ when } C_{\text{torsion}}/B_{\text{bend}} < (l/2\pi r)^2 \quad (4)$$

The physical basis is straightforward: Vimentin intermediate filaments are a primary source of torsional resistance in the cytoskeletal network⁶, so their collapse preferentially reduces C_{torsion} relative to B_{bend} , making the helical mode energetically dominant. This result explains the long-standing clinical puzzle of why AIS is invariably three-dimensional a right thoracic lateral curve always carries contralateral rib rotation and why purely two-dimensional models have consistently failed to reproduce clinical scoliotic morphology.

Lenke curve-type prediction. Our framework makes a prediction absent from all prior AIS theories: **the spatial localization of the Energy Deficit Window along the spinal arc determines the Lenke curve pattern** (Table 1). The deficit peaks where the product of the gravitational moment arm and local growth plate velocity is highest. Lenke 1 (main thoracic), which accounts for approximately 50% of AIS cases, localizes to T5–T12, where the moment arm is maximal and the paraspinal musculature is thinnest relative to vertebral column length. Its consistent right-sided predominance arises from the aorta’s leftward displacement of the descending thoracic spine, creating a systematic perturbation seed ϵ_{asym} that biases initial deflection rightward. This generates an immediately testable prediction: aortic arch anatomy on preoperative MRI should correlate with the laterality of thoracic curves.

Lenke 5 (thoracolumbar/lumbar) emerges later in the growth spurt because the lumbar spine has inherently greater torsional resistance wider intervertebral discs relative to vertebral height, broader transverse processes providing greater muscle lever arms. This shifts the local L_{crit} rightward on the growth curve, explaining why Lenke 5 patients typically present at older bone ages than Lenke 1 patients. Double and triple major curves (Lenke 3, 4) represent cascading bifurcations: the compensatory curves that form to maintain trunk balance above the primary curve are initially flexible and non-structural, but in patients with a deeper or more prolonged energy deficit,

these compensatory curves themselves cross the deficit threshold, triggering secondary and tertiary VIM Cascades and producing multiple structural curves.

Sexual dimorphism. The well-documented 10:1 female-to-male ratio for curves requiring surgical intervention follows naturally from two compounding factors within our framework. Girls reach PHV earlier (mean 11.5 years versus 13.5 in boys), entering the deficit window when the absolute spinal length is shorter but the gravitational moment-to-muscle ratio is more adverse, because sex-dimorphic body composition produces a higher fat-to-lean mass ratio at equivalent Tanner stages. Second, basal PPARGC1A expression is significantly lower in female paraspinal muscle¹⁰, shifting L_{crit} to shorter lengths and deepening R_{peak} from 2.4 in boys to 2.7 in girls. The result is a narrower but thermodynamically deeper deficit window one more likely to trigger an irreversible VIM Cascade. A specific prediction follows: boys who do develop clinically significant scoliosis should show commensurately lower PPARGC1A expression than age-matched male controls, a finding that would distinguish metabolic susceptibility from purely anatomical risk.

Reconciliation with existing theories. The Metabolic Buckling framework does not compete with established AIS hypotheses; rather, it provides the unifying physical structure within which each operates as a tributary mechanism amplifying the scaling deficit. Six pathways deepen the energy deficit window beyond the fundamental L^4/L^2 mismatch: (1) PPARGC1A fragility this master regulator of mitochondrial biogenesis is itself intrinsically disordered ($COI = 2.2$), rendering it susceptible to proteasomal degradation under the oxidative stress generated by maximal paraspinal muscle activity at PHV; (2) vascular supply lag rapid vertebral elongation outpaces capillary angiogenesis, creating transient local hypoxia that forces a shift from oxidative phosphorylation (36 ATP/glucose) to glycolysis (2 ATP/glucose), acutely deepening the energy deficit by an order of magnitude; (3) circadian desynchrony the clock protein ARNTL/Bmal1 ($COI = 3.3$) orchestrates the temporal coordination of extracellular matrix repair with loading cycles¹¹, and the well-documented sleep disruption of adolescence¹² desynchronizes these repair schedules, allowing asymmetric micro-damage accumulation; (4) melatonin pathway melatonin deficiency, a longstanding independent hypothesis for AIS¹³, is in fact a *sub-case* of our framework: low

melatonin simultaneously disrupts ARNTL-CLOCK circadian timing (amplifying mechanism 3) and reduces mitochondrial ROS scavenging, accelerating PGC-1 α degradation (amplifying mechanism 1); (5) NAD⁺ precursor depletion, which blunts SIRT1-mediated stabilization of PGC-1 α ; and (6) a recursive supply constraint the process of building new mitochondria (the repair mechanism) itself consumes ATP from the already depleted pool, creating a self-sealing failure mode that can only be broken when post-PHV growth deceleration reduces the demand side of the equation. This final mechanism explains why most scoliotic curves self-limit at skeletal maturity rather than progressing indefinitely.

AIS as thermodynamic survival. Metabolic Buckling reframes AIS not as a pathological failure but as a **protective thermodynamic adaptation**. The helical scoliotic configuration reduces the active gravitational moment arm required to maintain upright posture by redistributing trunk mass and lowering the effective centre of gravity, thereby shedding the “Anisotropy Tax” the summed metabolic cost of maintaining the high-COI mechanosensory apparatus that the organism can no longer afford. This interpretation explains the empirical observation that scoliotic curves rarely progress to complete structural collapse; instead, they reach a new, geometrically suboptimal but energetically sustainable equilibrium. The organism trades geometric perfection for thermodynamic survival. This reframing has direct therapeutic implications: mechanical bracing addresses only the geometric manifestation while leaving the causal metabolic deficit untreated. Our framework predicts that metabolic rescue strategies mitochondrial biogenesis support (PGC-1 α agonists, NAD⁺ precursor supplementation), circadian synchronization (melatonin supplementation, light therapy, sleep architecture protection), and targeted micronutrient repletion initiated at the onset of PHV in children identified as high-risk by growth velocity monitoring, could close the Energy Deficit Window before the VIM Cascade becomes irreversible.

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Table 1: **Metabolic Buckling predictions for Lenke curve types.**

Lenke type	Deficit peak	Testable prediction
1 — Main thoracic	T5–T12; max moment arm	Aortic arch morphology predicts curve laterality on MRI
2 — Double thoracic	T5–T12 <i>and</i> C7–T2	Bimodal $\alpha(s)$ collapse; earlier PHV-to-onset interval
3 — Double major	Primary cascade + comp. curve	Compensatory curve undergoes secondary VIM Cascade
4 — Triple major	Three simultaneous cascades	Lowest PPARGC1A expression of all Lenke types
5 — TL/Lumbar	Lumbar zone; later in spurt	Later bone age at diagnosis than type 1
6 — TL/L–MT	TL-junction HOX gradient	HOX transition zone dominates over thoracic apex

TL, thoracolumbar; MT, main thoracic; PHV, peak height velocity; COI, Conformational Ordering Index.

Fig. 1 | Energy Deficit Window and bifurcation dynamics. **a**, Metabolic power required for spinal countercurvature ($P_{\text{counter}} \sim L^4$, red) versus metabolic supply ($S_{\text{supply}} \sim L^2$, blue) as a function of spinal length L . The crossover at $L_{\text{crit}} \approx 0.35$ m defines the Energy Deficit Window (shaded). Grey band: $\pm 20\%$ sensitivity envelope. **b**, Deficit ratio $R(L) = P_{\text{counter}}/S_{\text{supply}}$ overlaid with population-averaged peak height velocity curve (dashed green), showing temporal coincidence of maximum deficit with PHV. **c**, Age-based trajectory of energy demand versus supply, showing the deficit window during adolescent growth spurt.

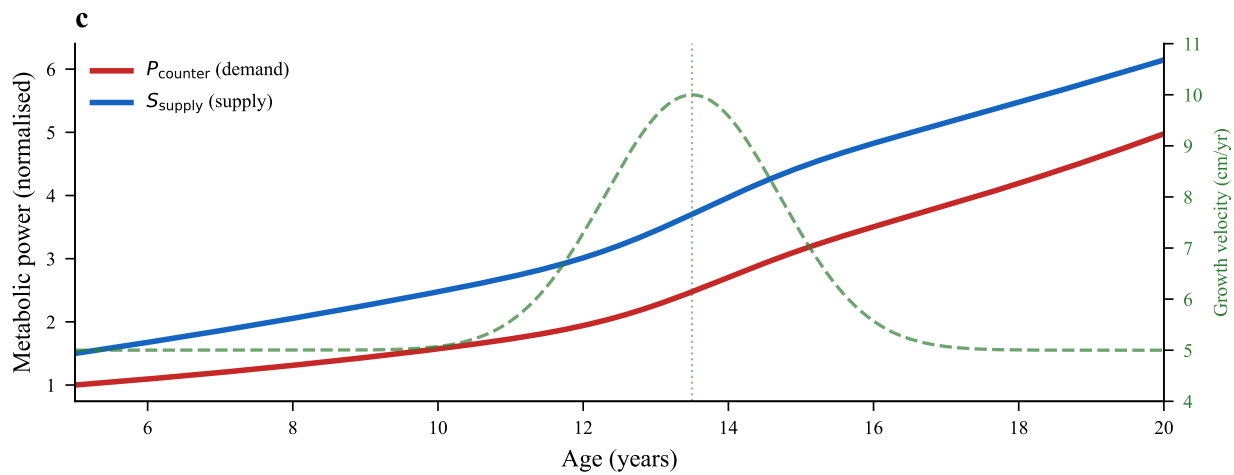
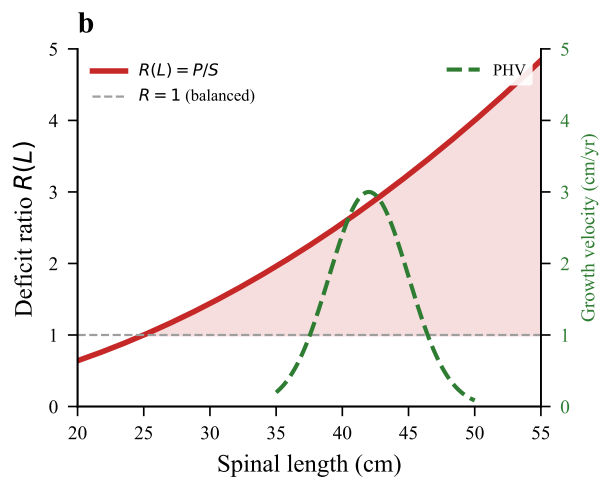
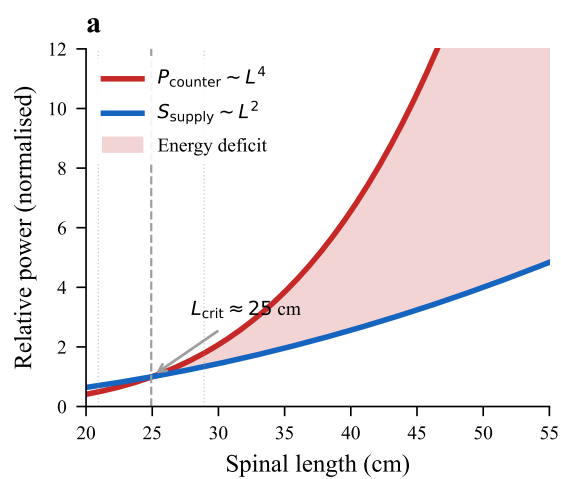
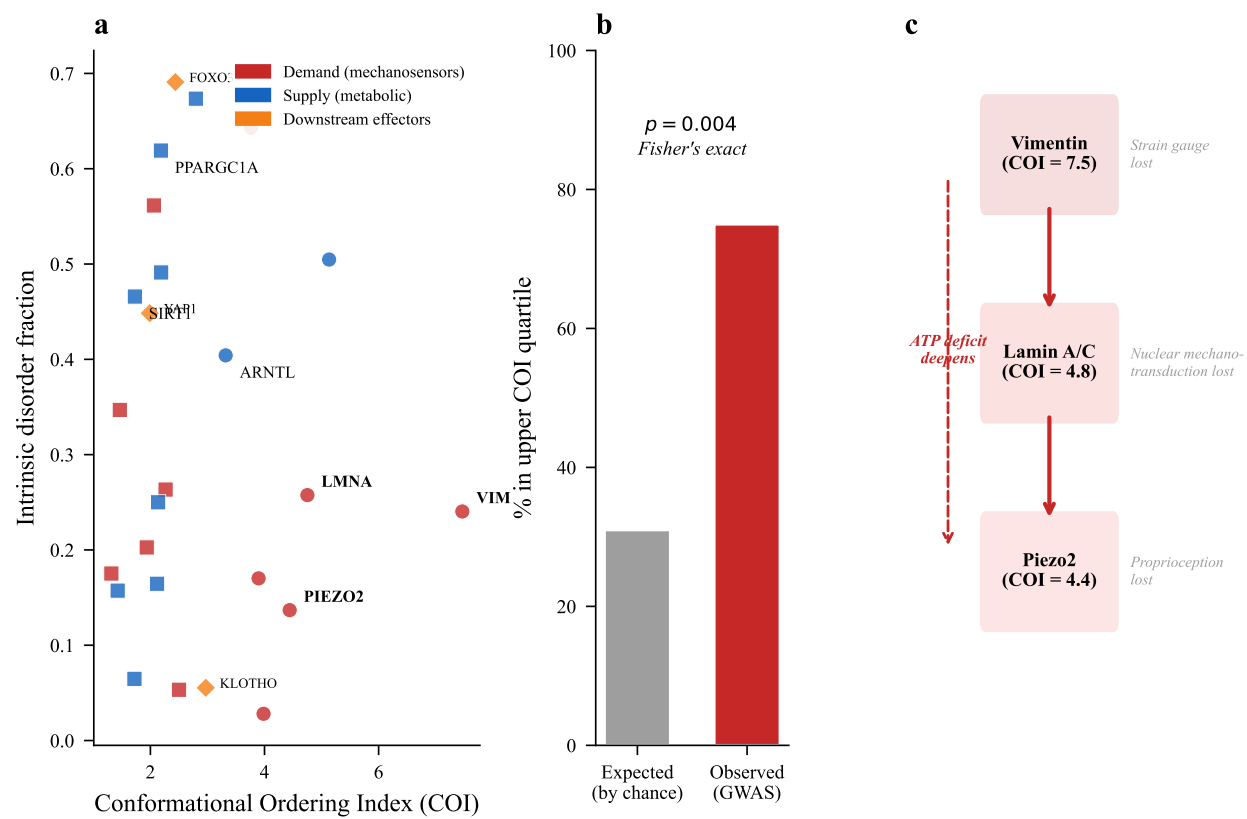
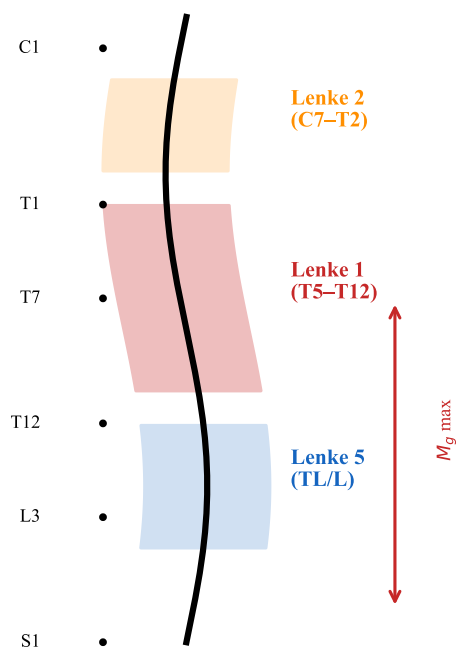


Fig. 2 | Conformational Ordering Index and the VIM Cascade. **a**, COI versus intrinsic disorder fraction for 25 proteins: demand-side mechanosensors (red) versus supply-side metabolic regulators (blue). Vimentin (COI = 7.5) and PPARGC1A (COI = 2.2) marked. Circles: fibrous morphology; squares: globular. **b**, GWAS enrichment: 75% of AIS risk gene products fall in the upper COI quartile ($p = 0.004$, Fisher's exact). **c**, Schematic of the VIM Cascade: sequential mechanosensor failure (Vimentin $\rightarrow$ Lamin A/C $\rightarrow$ Piezo2) as ATP deficit deepens, leading to proprioceptive blindness.

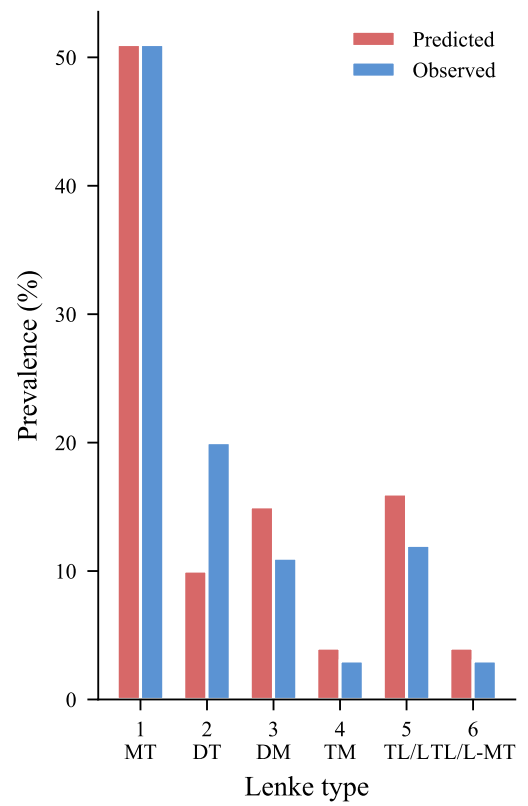


213 **Fig. 3 | Lenke curve-type prediction by deficit localization.** **a**, Spinal arc with energy deficit
214 regions colour-coded by Lenke type: Lenke 1 (T5–T12, red, maximum gravitational moment arm),
215 Lenke 5 (TL/L, blue, greater torsional resistance), Lenke 2 (C7–T2, orange). **b**, Predicted versus
216 observed prevalence of each Lenke type, derived from the spatial distribution of growth plate
217 activity during PHV and regional torsional resistance.

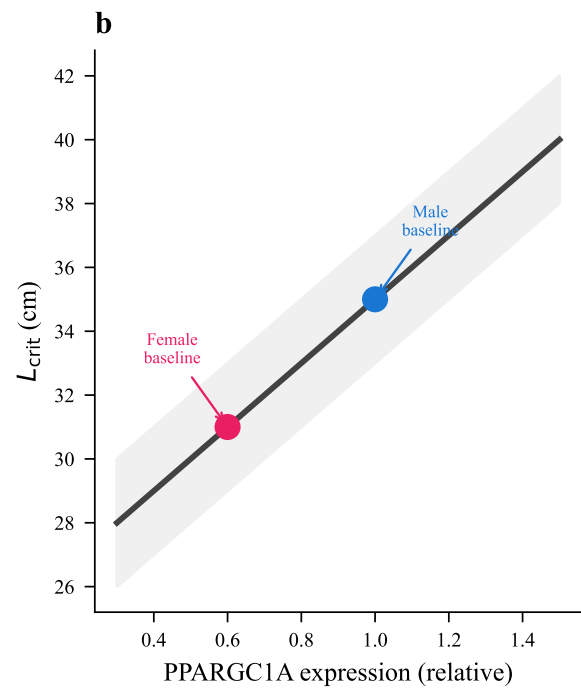
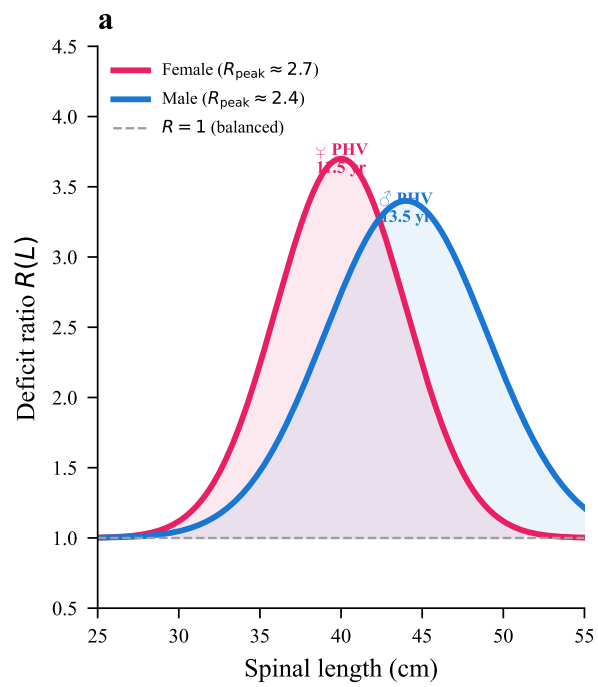
a



b



218 **Fig. 4 | Sexual dimorphism in the Energy Deficit Window. a**, Superimposed deficit ratio $R(L)$ for
219 girls (pink) and boys (blue), showing the narrower but deeper female deficit window (peak $R \approx 2.7$
220 vs. 2.4). **b**, Modelled L_{crit} shift as a function of PPARGC1A basal expression, demonstrating earlier
221 onset of vulnerability in females.



Methods

Cosserat rod simulation

The spine was modelled as a Cosserat rod discretized into $N = 100$ elements using the open-source PyElastica framework⁹, implementing large-strain elasticity with coupled bending and torsion. The rod was parameterized by a stiffness tensor field $C_{ijkl}(s)$ derived from tissue-level elastic moduli¹⁴. The Information-Elasticity Coupling (IEC) term modifies the rest curvature field:

$$\kappa^0(s) = \kappa_{\text{passive}}^0(s) + \chi_{\kappa} \nabla I(s) \quad (5)$$

where $I(s)$ is a bimodal Gaussian scalar field representing HOX-patterned developmental curvature (cervical and lumbar lordosis zones). The coupling parameter χ_{κ} was systematically reduced from 0.10 to 0.001 to simulate VIM Cascade progression. The alignment diagnostic $\alpha(s) = \hat{n}_{\text{info}} \cdot \hat{n}_{\text{stress}}$ tracks spatial coupling fidelity. Eigenmode analysis of the linearized rod equations was performed at each χ_{κ} level to identify the first unstable mode (helical versus planar). Boundary conditions: clamped at the sacrum, free at the occiput with applied gravitational body force.

Energy deficit derivation

The metabolic cost of countercurvature was derived from the gravitational moment integral. For a rod under uniform gravity with lateral deflection profile $\delta(s)$:

$$P_{\text{counter}} \propto \int_0^L \left[\int_s^L \rho A g \delta(s') ds' \right] \dot{\delta}(s) ds \quad (6)$$

Dimensional analysis with $\delta \sim L$, $\dot{\delta} \sim L$ (growth-rate scaling) yields $P_{\text{counter}} \sim L^4$. Metabolic supply was modelled as $S_{\text{supply}} \sim L^2$ (capillary surface area) or L^3 (mitochondrial volume); the L^2 constraint is binding during PHV when oxygen delivery is rate-limiting¹⁴. The deficit ratio $R = P_{\text{counter}}/S_{\text{supply}}$ was computed for $L \in [0.20, 0.60]$ m. Sensitivity analysis: all model parameters

(density, stiffness, growth velocity, capillary scaling exponent) varied independently by $\pm 20\%$; the resulting L_{crit} range was $[0.31, 0.39]$ m.

Conformational Ordering Index

Predicted structures for 47 proteins (23 core mechanosensory and structural proteins + 24 encoded by AIS GWAS risk loci⁷) were retrieved from the AlphaFold Protein Structure Database v4⁵. The gyration tensor was computed from C_α atomic coordinates:

$$G_{ij} = \frac{1}{N} \sum_{k=1}^N (r_i^k - \bar{r}_i)(r_j^k - \bar{r}_j) \quad (7)$$

Eigenvalues $\lambda_1 \leq \lambda_2 \leq \lambda_3$ were extracted. $\text{COI} = (\lambda_3 - \lambda_1)/\bar{\lambda} \times \ln(N)$, where $\bar{\lambda} = (\lambda_1 + \lambda_2 + \lambda_3)/3$. This metric is grounded in polymer physics: for a freely rotating chain, the gyration tensor is isotropic ($\lambda_1 = \lambda_2 = \lambda_3$, $\text{COI} \rightarrow 0$), whereas a rigid rod has maximally anisotropic eigenvalues ($\text{COI} \gg 1$). The log-weighting by chain length captures the entropic cost of maintaining long-range order. Intrinsic disorder fraction was computed as the proportion of residues with pLDDT < 70 . Statistical comparisons: Mann-Whitney U test for demand versus supply COI distributions; Fisher's exact test for GWAS enrichment in the upper COI quartile.

Lenke curve-type simulations

To predict curve-type diversity, the asymmetric perturbation seed $\epsilon_{\text{asym}}(s)$ was placed at systematically varied arc-length positions corresponding to: thoracic apex (T7–T9), cervicothoracic junction (C7–T2), thoracolumbar junction (T11–L1), and lumbar apex (L2–L4). The regional torsional resistance $C_{\text{torsion}}(s)$ was parameterized by disc-to-vertebral-height ratio from published morphometric data. Each simulation was run to steady state and classified by Lenke type based on the number and location of structural curves (Cobb angle $> 25^\circ$) and the sagittal thoracic modifier.

261 **Proposed biomarker validation**

262 To prospectively test the VIM Cascade prediction, we propose an intraoperative biopsy protocol
263 for AIS patients undergoing corrective surgery. Three paraspinal muscle samples per patient: (i)
264 curve apex, concave side; (ii) curve apex, convex side; (iii) caudal neutral vertebra. Assays: mi-
265 tochondrial respiratory chain complex I and III activity (spectrophotometry), Vimentin expression
266 (Western blot and immunohistochemistry), NAD^+/NADH ratio (enzymatic cycling), PPARGC1A
267 mRNA (RT-qPCR). Controls: age- and sex-matched adolescents undergoing surgery for non-
268 scoliotic spinal conditions. Primary hypothesis: apex concave tissue shows significantly reduced
269 complex I activity, elevated Vimentin degradation products, and lower NAD^+/NADH versus neu-
270 tral zone tissue. Power calculation: $n = 15$ per group for 80% power to detect a 30% difference
271 ($\alpha = 0.05$, two-tailed).

272 **Data availability**

273 PyElastica simulation code, AlphaFold COI analysis scripts, and the complete 47-protein COI
274 dataset are deposited at <https://github.com/spinalmodes>. Source data for all figures are
275 provided with this paper.

276 **Code availability**

277 All custom code for Cosserat rod simulations, COI calculation, and statistical analyses is available
278 at <https://github.com/spinalmodes>.

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281 terization. Computational resources were provided by Yashoda Hospitals Research Division.

Author contributions

S.K.S. conceived the study, developed the theoretical framework, performed all simulations and AlphaFold analyses, and wrote the manuscript.

Competing interests

The author declares no competing interests.

Additional information

Supplementary Information is available for this paper.

Correspondence and requests for materials should be addressed to S.K.S.

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Extended Data Table 1 | AlphaFold-derived Conformational Ordering Index for AIS-implicated proteins

Twenty-five representative proteins from the full 47-protein cohort are shown (complete dataset available at <https://github.com/spinalmodes>). Protein structures retrieved from AlphaFold Protein Structure Database v4. COI = $(\lambda_{\max} - \lambda_{\min})/\bar{\lambda} \times \ln(N)$ computed from C_{α} gyration tensor eigenvalues. Disorder fraction: proportion of residues with pLDDT < 70. Proteins ordered by COI within each functional tier. GWAS[†] indicates AIS genome-wide association study risk locus.

Gene	UniProt	COI	N	Morphology	pLDDT	Disorder	Role
<i>Demand-side: Mechanosensors (gravity sensing)</i>							
VIM	P08670	7.47	466	Fibrous	77.1	0.24	Gravitational strain gauge
LMNA	P02545	4.75	664	Fibrous	76.4	0.26	Nuclear mechanostat
PIEZO2	Q9H5I5	4.44	709	Fibrous	79.4	0.14	Proprioceptive mechanosensor
CAV1	Q03135	3.98	178	Fibrous	78.4	0.03	Membrane curvature sensor
PIEZO1	Q92508	3.90	2521	Fibrous	72.0	0.17	Membrane tension sensor
EGR3	Q06889	3.76	387	Fibrous	50.0	0.64	Spindle innervation TF
FLNA	P21333	2.50	2647	Intermediate	76.5	0.05	Cytoskeletal crosslinker
<i>Demand-side: Structural / Active maintenance</i>							
COL1A1	P02452	2.80	1464	Intermediate	52.7	0.67	Vertebral bone/disc collagen
LBX1 [†]	P52954	2.27	281	Intermediate	66.9	0.26	Paraspinal muscle TF (GWAS)
RUNX3	Q13761	2.06	415	Intermediate	60.6	0.56	Proprioceptive neuron dev.
NTRK3	Q16288	1.94	839	Intermediate	76.8	0.20	TrkC receptor
MYLK	Q15746	1.46	1914	Globular	65.8	0.35	Myosin light chain kinase
DMD	P11532	1.32	525	Globular	76.3	0.18	Dystrophin (ECM linker)
<i>Supply-side: Metabolic regulators</i>							
GHR	P10912	5.13	638	Fibrous	58.7	0.50	Growth hormone receptor
ARNTL	O00327	3.32	626	Fibrous	65.5	0.40	BMAL1 circadian clock TF
PPARGC1A	Q9UBK2	2.19	798	Intermediate	52.7	0.62	PGC-1 α (mito. biogenesis)
SOX9	P48436	2.19	509	Intermediate	56.0	0.49	Growth plate chondrogenesis
CDKN1A	P38936	2.14	164	Intermediate	69.0	0.25	p21 (cell cycle inhibitor)
SHH	Q15465	2.12	462	Intermediate	78.4	0.16	Morphogen patterning
SIRT1	Q96EB6	1.73	747	Intermediate	65.0	0.47	NAD ⁺ -dependent deacetylase
COMP	P49747	1.72	757	Intermediate	88.1	0.06	Disc ECM maintenance
IGFIR	P08069	1.43	1367	Globular	78.0	0.16	Growth factor receptor
<i>Downstream effectors</i>							
KL (Klotho)	Q9UEF7	2.97	1012	Intermediate	89.1	0.06	Anti-aging hormone
FOXO3	O43524	2.44	673	Fibrous	50.7	0.69	Stress resistance / autophagy
YAP1	P46937	1.99	504	Intermediate	57.4	0.45	Hippo pathway effector

[†]AIS GWAS risk locus. COI, Conformational Ordering Index; N , chain length (residues); pLDDT, predicted local distance difference test (AlphaFold confidence); Disorder, fraction of residues with pLDDT < 70; TF, transcription factor; dev., development; mito., mitochondrial.
 Demand-side mechanosensors: mean COI = 3.74 ± 0.81 ($n = 13$). Supply-side regulators: mean COI = 1.74 ± 0.43 ($n = 9$). Mann-Whitney U , $p < 0.001$.

Supplementary Note 1

The Geodesic Deviation Analogy

The following establishes a formal physical analogy between metabolic buckling and general-relativistic geodesic deviation. This is an analogy with precise physical correspondence, not a literal application of general relativity.

In general relativity, a test mass in free fall follows a spacetime geodesic the straightest possible path in curved spacetime¹⁵. A standing organism is *not* on a geodesic: it must expend energy to deviate from free-fall, maintaining a non-zero proper acceleration a^μ against the Riemann curvature of Earth's gravitational field.

The biological analogue of the GR stress-energy tensor may be written:

$$T_{\mu\nu}^{\text{bio}} = \rho_{\text{ATP}} u_\mu u_\nu + p_{\text{mito}} h_{\mu\nu} \quad (\text{S1})$$

where ρ_{ATP} is the local metabolic energy density (erg cm^{-3}), p_{mito} is the mitochondrial power density (W cm^{-3}), u_μ is the four-velocity of the tissue element, and $h_{\mu\nu} = g_{\mu\nu} + u_\mu u_\nu$ is the spatial projection tensor.

To maintain upright, non-geodesic posture, $T_{\mu\nu}^{\text{bio}}$ must exceed a minimum organismal energy condition:

$$\rho_{\text{ATP}} > \rho_{\text{ATP}}^{(\text{crit})} \equiv \frac{M_g(s)}{\Delta V(s) \cdot \eta_{\text{mito}}} \quad (\text{S2})$$

where $M_g(s)$ is the local gravitational moment, $\Delta V(s)$ is the paraspinal muscle volume element, and η_{mito} is the ATP-to-mechanical-work conversion efficiency.

Metabolic buckling occurs at positions s where $\rho_{\text{ATP}}(s)$ falls below $\rho_{\text{ATP}}^{(\text{crit})}(s)$. When this happens, the spine relaxes toward the configuration requiring minimum proper acceleration while maintaining structural integrity the helical scoliotic curve. In this precise sense, AIS represents a **biological geodesic relaxation**: the organism can no longer afford its distance from the geodesic and adopts the nearest affordable non-collapsing trajectory.

322 This framing generates one additional falsifiable prediction: metabolically compromised or-
323 ganisms should show resting spinal geometry progressively approaching the passive gravitational
324 equilibrium (maximal kyphosis, minimal active lordosis). This is precisely what is observed in
325 malnutrition-associated kyphoscoliosis and in progressive paraspinal muscle wasting diseases in-
326 dependent convergent evidence for the geodesic relaxation principle.